

Jacobsthal Products

All at Once

A Cantor-block construction and a finite-tail reduction

A research attempt on Altman's Question 2.16

20 September 2026

Abstract

Jacobsthal multiplication is obtained by transfinitely iterating natural addition. Its associativity is classical, but Altman asked for a direct interpretation of products of several, and even transfinitely many, factors without defining the answer by parenthesized iteration or a limit. We give a Cantor-normal-form-based construction on the actual Cartesian product in the finite case and on the finite-support Cartesian product in the transfinite case.

For finitely many factors, a tuple is assigned a pivot: the last coordinate, other than the first, lying in a positive-exponent Cantor block. The resulting finitely many cells receive ordinary reverse-lexicographic orders and are placed in decreasing order of their monomial types. A block comparison proves compatibility with finite regrouping without assuming Jacobsthal associativity. The construction is a componentwise linear extension and is maximal among orders subject to its within-cell constraints.

For arbitrary nonzero factors, ordinary and Jacobsthal partial products have the same Cantor exponents and the same leading monomial. A uniform bound $O_\xi \leq J_\xi < O_\xi \cdot 2$ implies equality at every limit with cofinally many nonunit factors. Removing units leaves an active index of the form $\lambda + n$, where λ is zero or a limit and n is finite. The entire product is an ordinary finite-support prefix followed by a single finite pivot product. We also obtain an exact three-state test for equality with the ordinary product and an order-theoretic proof of the binary natural-product upper bound. Exact computational checks accompany, but do not replace, the proofs.

Status. This is an unrefereed, AI-assisted research manuscript, not a priority certification or a proof-assistant formalization. The proposed answer uses a precise CNF-based meaning of “all at once.” The scope of the claim and the limitations of the literature search are stated explicitly in Sections 1 and 11.

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1 The selected problem and the proposed answer

1.1 A published question, rather than a newly invented conjecture

The selected problem is Question 2.16 in Harry Altman’s *Intermediate arithmetic operations on ordinal numbers* [1]. In the notation of this manuscript, it asks for an order-theoretic meaning of $a \times_J b \times_J c$ that does not first choose parentheses, and an interpretation of a transfinately indexed Jacobsthal product other than its recursive limit definition. The operation $\times_J$ is Altman’s $\times$; we reserve ordinary juxtaposition for ordinary ordinal multiplication.

The associativity identity itself is not an open problem. It is a classical result of Jacobsthal, reproduced in Altman’s Theorem 2.13. Nor is the binary Cantor-normal-form formula new. The objective here is to construct the underlying ordered objects directly and explain how the already known arithmetic emerges from them.

There is relevant prior work on infinitary natural sums and products. Lipparini’s work on transfinite natural sums provides related order-theoretic descriptions [4]; his countable natural-product construction combines finite natural-product behavior with ordinary finite-support products [5]. The ordinary finite-support model used below is standard, not a new construction. A recent general framework for noncommutative infinitary operations appears in [6]. Our proposed construction concerns Jacobsthal multiplication specifically.

A targeted literature search on 20 September 2026 located the published question and these related works, but did not locate a subsequent solution to Question 2.16. This is a statement about the search, not a proof of worldwide novelty. The mathematical arguments below can be assessed independently of that unresolved bibliographic question.

1.2 What “all at once” means here

We impose the following concrete requirements. The object must be defined on tuples, or on finite-support functions when the index is transfinite. Its comparison rule must use only the input well-orders and explicitly specified ordinary order constructions. Neither a parenthesized Jacobsthal product nor the supremum of its partial values may occur in the definition of that comparison rule. We also require invariance under order isomorphisms of the inputs and extension of the componentwise partial order.

These requirements permit the canonical Cantor decomposition of each input well-order. They do not require the comparison rule to be preserved by arbitrary order embeddings. They also do not require the usual flattening bijection between nested tuples to be the associativity isomorphism. These distinctions are substantive, not merely terminological.

1.3 The shape of the answer

For a finite list, we construct a well-order $\mathfrak{F}(A_1, \dots, A_r)$ on $A_1 \times \dots \times A_r$. Write $F(a_1, \dots, a_r)$ for its order type. Its definition, given in Section 3, is a single rule for all finite arities. Section 4 proves

$$\mathfrak{F}(U \cap V) \cong \mathfrak{F}(\mathfrak{F}(U), \mathfrak{F}(V)), \quad (1)$$

where U, V are finite lists of well-orders. This proof compares monomial cells and does not assume associativity of $\times_J$.

For a sequence $(a_i)_{i < \kappa}$ of nonzero ordinals, remove all factors 1 and let the remaining, or *active*, index have order type $\tau = \lambda + n$, with $\lambda = 0$ or λ a nonzero limit ordinal, and $n < \omega$. Denote the reindexed active factors by $(b_\eta)_{\eta < \tau}$. The main reduction is

$$J((a_i)_{i < \kappa}) = p F(b_\lambda, \dots, b_{\lambda+n-1}), \quad p = \prod_{\eta < \lambda} b_\eta. \quad (2)$$

Here the product defining p is *ordinary*; $p = 1$ when $\lambda = 0$. When $\lambda > 0$, p is a pure power of ω . The right side is realized by a direct order on finite-support functions: compare the finite terminal tuples using $\mathfrak{F}$; when they agree, compare the prefix functions by their largest differing coordinate. Thus the *definition of the final ordered set* contains no limit of Jacobsthal products.

Two further results make the reduction checkable and useful. At every partial stage ordinary and Jacobsthal products have identical CNF exponent supports and identical leading monomials. Moreover, equality of the two full products is decided by a three-state computation on the finite active suffix alone. Both statements are proved below for arbitrary ordinals, not merely for the computable ordinals used in the tests.

2 Conventions and the binary arithmetic

2.1 Cantor blocks, tails, and three multiplications

For a nonzero ordinal a , write

$$a = \omega^{\rho_1} c_1 + \dots + \omega^{\rho_s} c_s, \quad \rho_1 > \dots > \rho_s, \quad 0 < c_i < \omega. \quad (3)$$

Its degree is $d(a) = \rho_1$, its leading monomial is $\text{lm}(a) = \omega^{\rho_1} c_1$, and its CNF exponent support is $\text{supp}_{\text{CNF}}(a) = \{\rho_1, \dots, \rho_s\}$. A *monomial* has one distinct exponent, with any positive finite coefficient. A *pure power* has the more restrictive form ω^ρ . In particular, every positive finite ordinal is a monomial, but only 1 is a finite pure power.

There is a unique decomposition

$$a = a^+ + t(a), \quad t(a) < \omega, \quad (4)$$

where a^+ is zero or a nonzero limit ordinal. The number $t(a)$ is the coefficient of exponent 0, or zero if that exponent is absent. For a well-order A of type a , let T_A be its final finite interval of length $t(a)$.

Expand every coefficient in (3) into repeated unit blocks:

$$a = \omega^{\sigma_1} + \dots + \omega^{\sigma_m}, \quad \sigma_1 \geq \dots \geq \sigma_m. \quad (5)$$

Transporting the corresponding intervals along the unique isomorphism $A \cong a$ gives canonical convex blocks $B_{A,q}$ of type ω^{σ_q} . Blocks of exponent 0 are singletons and together form T_A . No choice of a noncanonical presentation is involved.

Natural sum $\oplus$ adds CNF coefficients at equal exponents. Natural product $\otimes$ multiplies the resulting formal polynomials, using natural addition for the exponents. In particular,

$$a \otimes n = \sum_{j=1}^s \omega^{\rho_j} (c_j n) \quad (n < \omega), \quad (6)$$

with zero terms omitted. In contrast, for $n > 0$,

$$an = \omega^{\rho_1}(c_1 n) + \omega^{\rho_2} c_2 + \cdots + \omega^{\rho_s} c_s. \quad (7)$$

Thus ordinary multiplication by a positive finite integer changes only the leading coefficient, whereas natural multiplication changes them all.

Jacobsthal multiplication is defined by

$$a \times_J 0 = 0, \quad a \times_J (\beta + 1) = (a \times_J \beta) \oplus a, \quad a \times_J \lambda = \sup_{\beta < \lambda} (a \times_J \beta) \quad (8)$$

at nonzero limit λ . We use these equations only to identify our constructed objects with the standard operation.

2.2 Ordinary absorption and a self-contained binary calculation

Lemma 2.1. *Let $a > 0$, $d = d(a)$, and $\sigma > 0$. Then*

$$a\omega^\sigma = \omega^{d+\sigma}. \quad (9)$$

For $b = b^+ + t$ as in (4),

$$ab = \omega^d b^+ + at. \quad (10)$$

Consequently $d(ab) = d(a) + d(b)$ when $a, b > 0$.

Proof. The equality $a\omega = \omega^{d+1}$ follows by taking the supremum of the finite multiples in (7). Induction on positive σ , using ordinary associativity at successors and continuity at limits, gives (9). Expanding b^+ into its positive-exponent CNF terms and using left distributivity of ordinary multiplication gives (10). Its highest exponent is $d(a) + d(b)$ when b is infinite. When b is finite and positive, the highest exponent remains $d(a) = d(a) + 0$. $\square$

Proposition 2.2 (Classical Jacobsthal formula). *For $a > 0$, $d = d(a)$, and $b = b^+ + t$, one has*

$$\boxed{a \times_J b = \omega^d b^+ + (a \otimes t).} \quad (11)$$

This is the formula in Jacobsthal's theorem, restated in [1, Theorem 2.10]. The proof below is included to isolate precisely the arithmetic input needed later.

Proof. Let $H(b)$ denote the right side of (11). Plainly $H(0) = 0$. All exponents in $\omega^d b^+$ are strictly greater than d . Therefore adding a naturally changes only the finite-tail contribution, and

$$H(b) \oplus a = \omega^d b^+ + (a \otimes (t + 1)) = H(b + 1).$$

Now let λ be a nonzero limit. Equations (6)–(10) imply $ab \leq H(b)$. For $b < \lambda$, every exponent in $a \otimes t(b)$ is at most d , so

$$H(b) < \omega^d b^+ + \omega^{d+1} = \omega^d (b^+ + \omega) \leq \omega^d \lambda.$$

The last inequality uses $b^+ + \omega \leq \lambda$, since λ is a limit larger than b . On the other hand, ordinary continuity gives

$$\sup_{b < \lambda} ab = a\lambda = \omega^d \lambda.$$

Hence $\sup_{b < \lambda} H(b) = \omega^d \lambda = H(\lambda)$. The defining recursion (8) proves the assertion. $\square$

Corollary 2.3. *For nonzero a, b ,*

$$a \times_J b = ab \iff t(b) \leq 1 \text{ or } a \text{ is a monomial.} \quad (12)$$

In particular, $a \times_J \omega^\sigma = a\omega^\sigma$ for $\sigma > 0$, and $\omega^\delta \times_J b = \omega^\delta b$ for every b .

Proof. The positive-exponent contributions in (10) and (11) coincide and lie above all exponents in the remaining terms. The two remaining terms agree when $t = 0$ or 1. For $t \geq 2$, comparison of (6) and (7) shows that they agree exactly when there are no lower CNF terms of a . $\square$

Remark 2.4. Ordinary addition in exponents is essential. It is strictly increasing in its right argument but need not be strictly increasing in its left argument. We repeatedly use $\delta + \sigma > \delta$ for $\sigma > 0$. We never replace the ordinary prefix-degree sums below by natural sums. All statements here make sense beyond ε_0 , even when an ordinal appears as its own leading exponent. Only the testing implementation is restricted to hereditary CNF expressions below ε_0 .

3 The finite pivot construction

3.1 A partition of the Cartesian product

Let $A_1, \dots, A_r$ be nonempty well-orders with types $a_1, \dots, a_r$, where $r \geq 1$. Write $B_{j,q}$ for the unit CNF blocks of A_j , with exponents $\sigma_{j,q}$, and put $T_j = T_{A_j}$. Set

$$D_0 = 0, \quad D_j = d(a_1) + \dots + d(a_j). \quad (13)$$

These are ordinary finite sums.

Definition 3.1 (Pivot and cell). For a tuple $x = (x_1, \dots, x_r)$, its pivot is the largest index $j \in \{2, \dots, r\}$ for which $x_j \notin T_j$, if such an index exists; otherwise its pivot is 1.

A cell with pivot $j \geq 2$ has the form

$$C(j, q; u_{j+1}, \dots, u_r) = A_1 \times \dots \times A_{j-1} \times B_{j,q} \times \{u_{j+1}\} \times \dots \times \{u_r\}, \quad (14)$$

where $\sigma_{j,q} > 0$ and $u_i \in T_i$. With pivot 1, the same formula means $B_{1,q} \times \{u_2\} \times \dots \times \{u_r\}$, and all blocks of A_1 , including exponent-zero blocks, are allowed.

The cells are disjoint and cover the product. Indeed, the pivot is unique, its block is unique, and every later coordinate is necessarily in its finite tail. There are finitely many cells: each factor has finitely many unit blocks and each T_i is finite. Some possible pivot indices contribute no cells because a required later tail is empty.

Within a cell use ordinary *colexicographic* order: compare the first j coordinates at the largest coordinate where they differ. This is the usual order model of an ordinary ordinal product. Define the exponent of the cell by

$$e(C) = D_{j-1} + \sigma_{j,q}. \quad (15)$$

For $j = 1$ this is simply $\sigma_{1,q}$.

Lemma 3.2. *Every cell, with its colexicographic order, has type $\omega^{e(C)}$.*

Proof. For $j = 1$ this is the definition of a unit block. For $j \geq 2$, the ordinary product of the first $j - 1$ factors has degree D_{j-1} by Lemma 2.1. Its product with $\omega^{\sigma_{j,q}}$ is $\omega^{D_{j-1} + \sigma_{j,q}}$, because the pivot exponent is positive. The fixed later coordinates do not affect the order type. $\square$

Lemma 3.3 (Separation of pivot layers). *If cells C and C' have pivots $j < k$, then $e(C) < e(C')$. For cells with the same pivot, comparison of their exponents $e(C)$ is exactly comparison of their input block exponents $\sigma_{j,q}$.*

Proof. For the first assertion,

$$e(C) \leq D_j \leq D_{k-1} < D_{k-1} + \sigma_{k,q'} = e(C'),$$

because $k \geq 2$ and its pivot block has positive exponent. The second assertion is strict monotonicity of addition by a fixed ordinal on the left. In particular, cell exponents from different pivot layers never collide, even when ordinary exponent addition absorbs a prefix. $\square$

3.2 The comparison rule

Definition 3.4 (Finite all-at-once order). Order the cells first by *decreasing pivot index j* , then by *decreasing input block exponent $\sigma_{j,q}$* . For ties, order by the colexicographic order of the finite tuple $(u_{j+1}, \dots, u_r)$, then by increasing block index q . Inside each cell retain its colexicographic order. Denote the concatenation by $\mathfrak{F}(A_1, \dots, A_r)$ and its type by $F(a_1, \dots, a_r)$.

For the empty list put $\mathfrak{F}(\langle \rangle) = \mathbf{1}$, a singleton. If a factor is empty, put $\mathfrak{F}(A_1, \dots, A_r) = \emptyset$.

The order is a well-order because it is a finite concatenation of well-orders. By Lemma 3.3, it also lists the cells in decreasing order of their monomial exponents $e(C)$. The tie rule is deterministic; different reasonable tie rules among equal monomial types need not give the same pointwise order, but give the same order type. The chosen rule agrees with ordinary colex when all factors are finite. For a single factor it gives its original well-order.

Most importantly, Definition 3.4 never evaluates a Jacobsthal product. It uses canonical convex blocks, finite tags, and ordinary colexicographic comparison. The prefix-degree sums D_j are needed only to *prove the resulting order type*, not to compare tuples. Thus the comparison rule is even more elementary than a rule that first computes all the monomial types, and is not a recursive definition disguised as an r -ary notation.

Theorem 3.5 (Componentwise extension). *The order $\mathfrak{F}(A_1, \dots, A_r)$ extends the componentwise partial order on $A_1 \times \dots \times A_r$.*

Proof. Suppose $x_i \leq y_i$ for every i , and let j_x, j_y be the pivots. A coordinate in a final finite interval cannot leave that interval by increasing. Hence $j_y \leq j_x$.

First suppose $j_y < j_x$. Since $j_x \geq 2$, its pivot exponent is positive. The exponent of the cell of y satisfies

$$e(C_y) \leq D_{j_y} \leq D_{j_x-1} < D_{j_x-1} + \sigma_{j_x,q_x} = e(C_x).$$

Here the first inequality also covers $j_y = 1$. The cell of x is therefore placed before that of y .

Now suppose $j_x = j_y = j$. Increasing x_j cannot decrease its block index, and so cannot increase its block exponent. Strict monotonicity of ordinary addition in the right argument gives $e(C_y) \leq e(C_x)$. A strict inequality again puts x first. In the equality case the pivot exponents are equal. The later

tail tuples are coordinatewise ordered, and hence are ordered in their colexicographic tie rule. If they agree, the block index of x_j is no larger than that of y_j . Finally, if both tuples lie in the same cell, its internal colex order respects every coordinatewise comparison. These cases prove $x \leq_{\mathfrak{F}} y$. $\square$

3.3 The explicit all-arity formula

For $j \leq r$, write $T_{j,r} = \prod_{i=j}^r t(a_i)$, an ordinary finite integer product, and set $T_{r+1,r} = 1$. Terms with coefficient zero are omitted.

Theorem 3.6 (Cell formula). *For nonzero $a_1, \dots, a_r$,*

$$F(a_1, \dots, a_r) = a_1 \otimes T_{2,r} \oplus \bigoplus_{j=2}^r \bigoplus_{\substack{(\sigma,c) \in \text{CNF}(a_j) \\ \sigma > 0}} \omega^{D_{j-1} + \sigma}(cT_{j+1,r}). \quad (16)$$

All natural sums displayed here are finite.

Proof. Every unit block of a_1 gives one pivot-1 cell for each choice of later tail coordinates, hence $T_{2,r}$ copies of its monomial. Every positive-exponent unit block of a_j gives $T_{j+1,r}$ cells of degree $D_{j-1} + \sigma$. Lemma 3.2 identifies the types. Sorting the finitely many monomials in decreasing exponent produces their natural sum. Collecting repeated unit blocks yields (16). $\square$

Corollary 3.7 (Binary identification). *For all ordinals a, b ,*

$$F(a, b) = a \times_J b.$$

Proof. Zero factors are immediate. The positive blocks of b contribute $\omega^{d(a)}b^+$. Their exponents are strictly greater than $d(a)$, so they precede every term of the contribution $a \otimes t(b)$. Therefore the cell formula is exactly (11). $\square$

Lemma 3.8 (Degree and terminal cardinality). *For a nonempty finite list of nonzero factors,*

$$d(F(a_1, \dots, a_r)) = D_r, \quad t(F(a_1, \dots, a_r)) = \prod_{i=1}^r t(a_i). \quad (17)$$

The final finite interval of $\mathfrak{F}$ consists precisely of the tuples all of whose coordinates belong to their final finite intervals.

Proof. If all factors are finite, the claims follow directly. Otherwise let h be the last index with $d(a_h) > 0$. All later factors are positive finite ordinals, so a cell using the leading block at h exists. Its exponent is $D_{h-1} + d(a_h) = D_r$. Every other cell has exponent at most D_r , which proves the degree assertion. This argument includes $h = 1$.

A pivot at $j \geq 2$ always has positive exponent. Thus exponent-zero cells arise exactly from an exponent-zero block at index 1 and choices in all later finite tails. They are singletons, and their number is the stated product. Every positive-exponent cell precedes them. $\square$

4 Regrouping and a restricted maximality principle

4.1 Why associativity follows from the construction

Theorem 4.1 (Finite concatenation). *For finite lists U, V of well-orders,*

$$\mathfrak{F}(U \smallfrown V) \cong \mathfrak{F}(\mathfrak{F}(U), \mathfrak{F}(V)). \quad (18)$$

The isomorphism is the unique order isomorphism between the two well-orders. The proof does not use Jacobsthal associativity.

Proof. Empty factors give empty orders on both sides. An empty list gives a singleton, which is a unit for the binary construction by its definition or by (11). We may therefore suppose both lists are nonempty and all factors are nonzero.

Let δ_U be the ordinary sum of degrees in U . By Lemma 3.8, it is the degree of $\mathfrak{F}(U)$. The canonical unit CNF blocks of $\mathfrak{F}(V)$ are precisely its cells, listed in the order of their exponents and tags: each such cell has a pure monomial type. We compare the monomial cells obtained on the right with those on the left in two groups.

A positive-exponent cell of $\mathfrak{F}(V)$ has some exponent $\eta > 0$. In the outer binary construction, it contributes one cell of type $\omega^{\delta_U + \eta}$. If its pivot was the j th coordinate of V , then

$$\eta = D_{j-1}^V + \sigma_{j,q}, \quad \delta_U + \eta = (\delta_U + D_{j-1}^V) + \sigma_{j,q}.$$

The second expression is exactly the exponent of the corresponding global cell whose pivot is in V . Its later tail coordinates have not changed. When the local pivot is 1, positivity means that its first block has positive exponent, as required for a global pivot beyond the first coordinate. Thus this correspondence preserves all multiplicities.

Next consider the exponent-zero cells of $\mathfrak{F}(V)$. By Lemma 3.8, these are its terminal points, in bijection with tuples of terminal points of the factors in V . For each such point, the outer construction takes every unit block of $\mathfrak{F}(U)$. These are exactly the cells of $\mathfrak{F}(U)$, and their exponents remain unchanged. They match the global cells whose pivot is in U , with all V -coordinates terminal.

We have matched the two finite multisets of pure monomial cell types. Each construction sorts this multiset in decreasing exponent. Consequently the ordered runs of equal exponents have the same finite lengths. Match the first block of each run on one side to the first block on the other, and so on. Inside matched blocks take the unique isomorphism of well-orders of type ω^η . Their union is the required order isomorphism. $\square$

Corollary 4.2. *For every finite list, $F(a_1, \dots, a_r)$ equals every parenthesized Jacobsthal product of its entries. In particular, $\times_J$ is associative. All coherence diagrams made from the order isomorphisms in Theorem 4.1 commute.*

Proof. Use Corollary 3.7 at each binary vertex and Theorem 4.1 to identify the result with the common all-arity object. A well-order has at most one order isomorphism to a given well-order, so two composites with the same endpoints coincide. $\square$

Remark 4.3 (The flattening bijection is not the associator). The assertion is an order isomorphism, not equality of comparison rules under naive tuple flattening. Take factors $\omega + 1, 2, \omega$. The direct three-factor construction has a single pivot-3 cell, ordered by ordinary colex. It places $(\omega, 0, 0)$

before $(0, 1, 0)$. In the nested order $\mathfrak{F}(\mathfrak{F}(\omega + 1, 2), \omega)$, the inner order places $(0, 1)$ before $(\omega, 0)$, reversing that comparison. Both total orders have type ω^2 ; their unique isomorphism is not the flattening identity. This is precisely why the block-isomorphism argument is needed.

4.2 A maximum without building the target value into the definition

The preceding definition selects an order. There is also an extremal characterization which removes the significance of its particular tie rule.

Lemma 4.4 (Finite interleaving bound). *Let a set be partitioned into finitely many well-orders $C_1, \dots, C_m$ of types $\alpha_1, \dots, \alpha_m$. Every linear order on their union whose restrictions are the given cell orders is a well-order of type at most $\alpha_1 \oplus \dots \oplus \alpha_m$.*

Proof. For a point z and a cell C_i , let $r_i(z)$ be the order type of the points of C_i preceding z in the proposed linear order. This is an initial segment of C_i , so $r_i(z) \leq \alpha_i$. If $z < z'$, all these ranks are nondecreasing, and the rank of the cell containing z strictly increases: that cell counts z before z' but not before itself. Strict monotonicity of finite natural sum therefore makes

$$z \mapsto r_1(z) \oplus \dots \oplus r_m(z)$$

strictly increasing. Its value is strictly below $\alpha_1 \oplus \dots \oplus \alpha_m$, because at z the rank in its own cell is strictly smaller than the full cell type. Thus it embeds the linear order into the indicated ordinal, proving both assertions. $\square$

Theorem 4.5 (Pivot-cell maximality). *Among linear extensions of the componentwise order on $A_1 \times \dots \times A_r$ whose restriction to every pivot cell is ordinary colexicographic order, $\mathfrak{F}(A_1, \dots, A_r)$ has the largest possible order type.*

Proof. By Theorem 3.5, $\mathfrak{F}$ is an admissible order. Every admissible order is bounded by the finite natural sum of the cell types, by Lemma 4.4. Since those types are pure powers, concatenating them in decreasing exponent attains that bound. This is exactly $\mathfrak{F}$. $\square$

The constraints in this theorem refer only to the input CNF blocks and the ordinary orders inside explicitly described subsets. They do not refer to $\times_J$ or to the desired order type. The result is therefore more than the tautology that a computed ordinal can order a set of suitable cardinality. It remains a *restricted* maximality theorem: if the within-cell restrictions are removed, the maximum is generally larger.

4.3 An order-theoretic comparison with natural multiplication

Corollary 4.6. *For every finite list,*

$$F(a_1, \dots, a_r) \leq a_1 \otimes \dots \otimes a_r.$$

In particular, $a \times_J b \leq a \otimes b$ has an order-theoretic proof.

Proof. Carruth's theorem identifies the natural product of finitely many ordinals as the maximum order type of a linear extension of their componentwise Cartesian product [3]; see also the explicit statement in [5, Section 3.3]. Our construction is one such extension, by Theorem 3.5. Zero factors are immediate. $\square$

The numerical inequality is already known. What this corollary supplies is the interpretation requested for the *binary multiplication part* of Altman's Question 5.2 [1]. It is not a claimed solution to all the exponentiation and infinitary comparison questions grouped there.

5 Synchronization of ordinary and Jacobsthal products

5.1 The common-support invariant

Fix a sequence $(a_i)_{i < \kappa}$ of *nonzero* ordinals. Define ordinary and Jacobsthal partial products by

$$\begin{aligned} O_0 &= J_0 = 1, \\ O_{\xi+1} &= O_\xi a_\xi, & J_{\xi+1} &= J_\xi \times_J a_\xi, \end{aligned} \tag{19}$$

and at nonzero limits use the suprema of the preceding partial products. All factors are at least 1, so both sequences are nondecreasing. The treatment of a zero factor is separated in Section 7.3; a supremum formula for nondecreasing products must not be used carelessly after multiplication by zero.

Lemma 5.1 (One-step synchronization). *Suppose $x, y > 0$ have the same CNF exponent support and leading monomial, and every CNF coefficient of x is at most the corresponding coefficient of y . For every $b > 0$, the pair $xb, y \times_J b$ has the same properties. If $t(b) = 0$, then $xb = y \times_J b$.*

Proof. Let $d = d(x) = d(y)$ and $b = b^+ + t$. The two outputs are

$$\omega^d b^+ + xt, \quad \omega^d b^+ + (y \otimes t).$$

When $t = 0$ they coincide. When $t > 0$, the terms contributed by b^+ are identical and have exponents strictly above d . In the remaining terms, xt multiplies the common leading coefficient by t and leaves all lower coefficients of x unchanged. The ordinal $y \otimes t$ multiplies every coefficient of y by t . Thus the remaining exponent supports are identical and all coefficient inequalities are preserved. If $b^+ \neq 0$, the leading monomial comes from the identical upper part. If $b^+ = 0$, it comes from the identically scaled old leading monomial. $\square$

Lemma 5.2 (A uniform one-step-scale bound). *If $x, y > 0$ have the same leading monomial and $x \leq y$, then*

$$x \leq y < x \cdot 2.$$

Proof. Write their leading monomial as $\omega^d c$, with $c \geq 1$ finite. Both ordinals are less than $\omega^d(c+1)$, while

$$\omega^d(c+1) \leq \omega^d(2c) \leq x \cdot 2.$$

The first of these inequalities also covers $d = 0$. The result follows. $\square$

Theorem 5.3 (Synchronization at every ordinal stage). *For every $\xi \leq \kappa$,*

$$\text{supp}_{\text{CNF}}(O_\xi) = \text{supp}_{\text{CNF}}(J_\xi), \quad \text{lm}(O_\xi) = \text{lm}(J_\xi). \tag{20}$$

Every coefficient of O_ξ is at most its corresponding coefficient in J_ξ , and consequently

$$\boxed{O_\xi \leq J_\xi < O_\xi \cdot 2.} \tag{21}$$

If $\lambda \leq \kappa$ is a nonzero limit and $\{i < \lambda : a_i > 1\}$ is cofinal in λ , then

$$O_\lambda = J_\lambda. \tag{22}$$

Proof. We prove all the assertions simultaneously by transfinite induction. They hold at stage 0. The successor step follows from Lemma 5.1; coefficient domination implies ordinal domination, and Lemma 5.2 gives the strict upper bound.

Let λ be a nonzero limit. If the indices $i < \lambda$ with $a_i > 1$ are bounded, choose a stage below λ after all of them. Every later factor is 1, so both product sequences are constant from that stage onward. Their limit values retain all the inductive properties.

Otherwise those indices are cofinal. For each $\xi < \lambda$, choose i with $\xi \leq i < \lambda$ and $a_i > 1$. Since λ is a limit, $i + 1 < \lambda$. The inductive bound and monotonicity give

$$J_\xi < O_\xi \cdot 2 \leq O_i \cdot 2 \leq O_i a_i = O_{i+1} \leq O_\lambda. \quad (23)$$

Taking the supremum over ξ yields $J_\lambda \leq O_\lambda$. The reverse inequality follows from $O_\xi \leq J_\xi$ at earlier stages. Hence the limit values agree. Their supports, leading monomials, and coefficients therefore agree as well; the strict factor-two bound is immediate for a positive ordinal compared with itself. This completes the induction. $\square$

Leading-monomial estimates also play a role in the natural-product analysis in [5, Lemmas 2.4–2.5]. Here the invariant compares *ordinary* and *Jacobsthal* partial products at arbitrary ordinal stages.

The critical point is (23). It does not presume that natural sum or Jacobsthal multiplication is continuous in an inappropriate argument. A later nonunit ordinary factor overtakes any current Jacobsthal value, because the discrepancy never reaches the next doubling of the leading scale.

5.2 Why the cofinally active limit is a pure power

Proposition 5.4. *Under the cofinality hypothesis of (22), the common value $p = O_\lambda = J_\lambda$ has the form ω^δ with $\delta > 0$.*

Proof. Every partial product is strictly below p . Indeed, after any stage there is a nonunit factor, which makes a subsequent ordinary partial product strictly larger. Given $u, v < p$, select $\xi < \lambda$ with $u, v < O_\xi$. Choose a later nonunit factor as in the preceding proof. Then

$$u + v < O_\xi \cdot 2 \leq O_{i+1} < p.$$

Thus p is additively indecomposable. For completeness, the CNF criterion for a positive ordinal to have this property is that it be a pure power of ω : if its leading coefficient is at least two, or if it has a nonzero lower tail, one can split off a leading monomial to express it as the sum of two smaller ordinals. Conversely, ordinals below a pure power have sum below that power, by ordinary CNF addition.

Cofinitely many stages are not needed, but cofinally many nonunit stages ensure that the value exceeds every finite integer: choose arbitrarily many such stages and compare with the corresponding product of 2's. Hence $p \geq \omega$, giving $\delta > 0$. $\square$

Example 5.5 (A limit index by itself is insufficient). Let $a_0 = \omega + 1$, $a_1 = 2$, and $a_i = 1$ for $2 \leq i < \omega$. Then

$$O_\omega = \omega \cdot 2 + 1, \quad J_\omega = \omega \cdot 2 + 2.$$

Both sequences stabilize after the second factor. The ambient index is a limit, but the active index is the finite ordinal 2.

Remark 5.6. Finite degree sums do not directly extend to all transfinite stages. For example, every factor in the constant sequence $2, 2, \dots$ has degree zero, but its ordinary and Jacobsthal products at ω are both ω , of degree one. The colexicographic prefix construction avoids any need for a false transfinite extension of (17).

6 Removing units and isolating the finite active tail

6.1 The active index

Definition 6.1. For a nonzero sequence $(a_i)_{i < \kappa}$ let

$$S = \{i < \kappa : a_i > 1\}.$$

Give S its inherited well-order, write $\tau = \text{otp}(S)$, and let $e : \tau \rightarrow S$ be its increasing enumeration. Put $b_\eta = a_{e(\eta)}$. Write uniquely

$$\tau = \lambda + n, \quad n < \omega, \quad (24)$$

where λ is zero or a nonzero limit. The factors indexed by $\lambda, \dots, \lambda + n - 1$ form the *finite active suffix*.

The decomposition (24) is simply the removal of the finite terminal part of the ordinal τ . It is not the assertion that all but finitely many original factors equal 1. There may be arbitrarily many nonunit factors in the initial part λ .

Lemma 6.2 (Unit erasure). *Deleting all factors equal to 1 does not change either the ordinary or the Jacobsthal product.*

Proof. For $\xi \leq \kappa$, let $s(\xi) = \text{otp}(S \cap \xi)$. Compare the original partial product at ξ with the active partial product at $s(\xi)$. Induction on ξ proves equality for each of the two operations. At a successor, either the new factor is 1 and neither active count nor value changes, or the new active factor is appended on both sides. At a limit, if the active indices below it are bounded, the values stabilize after the last relevant initial segment. Otherwise their counts are cofinal in the limit active count, and the two suprema are taken over cofinal copies of the same nondecreasing sequence. This includes the possibility of infinitely many active indices in a bounded initial segment; after that segment the value is simply constant. $\square$

Theorem 6.3 (Finite-tail reduction). *With the above notation, let*

$$p = \prod_{\eta < \lambda} b_\eta$$

be the ordinary product, with $p = 1$ when $\lambda = 0$. Then

$$\begin{aligned} J((a_i)_{i < \kappa}) &= p F(b_\lambda, \dots, b_{\lambda+n-1}), \\ O((a_i)_{i < \kappa}) &= p (b_\lambda \cdots b_{\lambda+n-1}). \end{aligned} \quad (25)$$

When $\lambda > 0$, $p = \omega^\delta$ for some $\delta > 0$. Empty suffixes have value 1.

Proof. First erase units. If $\lambda = 0$, the first identity is the finite identification in Corollary 4.2; the second is the definition of a finite ordinary product.

Suppose $\lambda > 0$. Every active factor is greater than 1, so Theorem 5.3 and Proposition 5.4 give $J_\lambda = O_\lambda = p = \omega^\delta$. Only n successor stages remain. Finite associativity, already proved from the cell construction, yields

$$J_{\lambda+n} = p \times_J F(b_\lambda, \dots, b_{\lambda+n-1}) = p F(b_\lambda, \dots, b_{\lambda+n-1}).$$

The last equality follows from Corollary 2.3, because p is a monomial. Ordinary associativity gives the second identity. $\square$

Corollary 6.4. *The full ordinary and Jacobsthal products are equal if and only if their finite active suffixes have equal ordinary and Jacobsthal products.*

Proof. Left multiplication by a fixed nonzero ordinal is strictly increasing in the right operand. The common nonzero factor p in (25) can therefore be cancelled for equality. $\square$

The reduction is stronger than a numerical estimate: all possible Jacobsthal-specific behavior can be moved into one finite product at the end of the active ordering. The prefix need not be small, countable, or computably presented.

7 A direct model for every ordinal-indexed product

7.1 The ordinary finite-support prefix

Let $(A_i)_{i < \theta}$ be nonempty well-orders, each with its least element denoted 0_i . Define

$$\text{FS}((A_i)_{i < \theta}) = \left\{ f \in \prod_{i < \theta} A_i : \{i : f(i) \neq 0_i\} \text{ is finite} \right\}. \quad (26)$$

For distinct f, g , their finite union of supports contains a largest coordinate at which they differ. Put $f <_{\mathfrak{C}} g$ when $f(i) < g(i)$ at that coordinate. Write $\mathfrak{C}((A_i)_{i < \theta})$ for this ordered set. The comparison rule is defined simultaneously on the full set (26).

Lemma 7.1 (Standard finite-support model). *The order $\mathfrak{C}((A_i)_{i < \theta})$ is a well-order extending the componentwise partial order. Its type is the ordinary ordinal product $\prod_{i < \theta} \text{otp}(A_i)$.*

Proof. The componentwise assertion follows immediately from the largest-difference rule. For the type and well-order assertions, induct on θ . At zero the set is a singleton. At a successor, it is the ordinary colex product of the previous finite-support order and the last factor, so its type is multiplied on the right by that factor's type.

At a limit, every finite support is contained in some proper initial segment of the index. The orders on these initial segments, extended by zero on later coordinates, form an increasing family of *initial suborders*: any function with a nonzero later coordinate is larger than all functions supported in the earlier segment. Their union is therefore a well-order of type the supremum of their types. This proves the recursion for the ordinary product. It is a proof of the direct model, not its definition. $\square$

This construction is recalled, with references to earlier sources, in [5, Section 3.2]. Finite support is essential. For example, the full Cartesian product of countably many copies of 2 has uncountable cardinality, whereas the ordinary product 2^ω is ω .

7.2 The final comparison rule

Return to the original nonzero factors and their active enumeration. Use the corresponding well-orders B_η for b_η . Define

$$\mathfrak{B} = \mathfrak{C}((B_\eta)_{\eta < \lambda}), \quad \mathfrak{T} = \mathfrak{F}(B_\lambda, \dots, B_{\lambda+n-1}).$$

When a list or prefix is empty, the associated order is a singleton.

Definition 7.2 (The transfinite all-at-once order). On $\text{FS}((B_\eta)_{\eta < \tau})$, split a function into its prefix f^- on λ and its finite terminal tuple f^+ . Put

$$f <_{\mathfrak{L}} g \iff (f^+ <_{\mathfrak{T}} g^+) \text{ or } (f^+ = g^+ \text{ and } f^- <_{\mathfrak{B}} g^-). \quad (27)$$

Restore the original unit coordinates, each of which has only its least point, to regard this as an order on $\text{FS}((A_i)_{i < \kappa})$. Denote it by $\mathfrak{L}((A_i)_{i < \kappa})$.

This is simply the ordinary colex product $\text{Cox}(\mathfrak{B}, \mathfrak{T})$, with the terminal tuple the more significant coordinate. No ordinal value J_ξ , no iteration of $\times_J$, and no supremum occurs in the comparison rule. Even the order type of $\mathfrak{B}$ need not be calculated to compare two given prefix functions.

Theorem 7.3 (All-at-once representation). *For every set-sized ordinal-indexed family of nonempty well-orders, $\mathfrak{L}$ is a well-order extending the componentwise order, and*

$$\text{otp}(\mathfrak{L}((A_i)_{i < \kappa})) = J(\text{otp}(A_i)_{i < \kappa}). \quad (28)$$

The construction is invariant under order isomorphisms of the index and of all factors.

Proof. By Lemma 7.1, $\mathfrak{B}$ is a well-order of type p . The finite order $\mathfrak{T}$ has type $F(b_\lambda, \dots, b_{\lambda+n-1})$. Their ordinary colex product is a well-order of the product of these types. Theorem 6.3 identifies this type with the desired Jacobsthal product.

If $f \leq g$ componentwise, their suffixes are componentwise ordered. If the suffixes differ, Theorem 3.5 compares them in the required direction. If they agree, Lemma 7.1 compares the prefixes in that direction. Thus (27) extends the full componentwise order. Restoring unit coordinates is a bijection preserving that partial order.

Every operation in the definition is preserved by isomorphisms: the active index, its finite terminal part, the canonical CNF blocks, their exponents, the least points, finite support, and largest-coordinate comparison. This proves invariance. $\square$

Corollary 7.4 (Finite-cell maximality in the transfinite case). *If the active suffix is nonempty, partition its tuple set into its finite pivot cells C . Partition the full finite-support set into the corresponding subsets $\mathfrak{B} \times C$, ordered internally by ordinary colex. Among componentwise linear extensions having these specified within-cell orders, $\mathfrak{L}$ has the largest order type. If the suffix is empty, its one prescribed cell is the ordinary prefix order itself.*

Proof. The type of $\mathfrak{B}$ is a pure power ω^δ , allowing $\delta = 0$ when the prefix is empty. A suffix cell of type ω^e gives a full cell of type $\omega^{\delta+e}$. Since $e \mapsto \delta + e$ is strictly increasing, the decreasing-exponent order of the suffix cells also sorts the full cells correctly. The same finite interleaving bound (Lemma 4.4) proves maximality. $\square$

7.3 Zero factors and the empty product

The empty product is a singleton of type 1. If some factor is empty, the corresponding Cartesian set of finite-support functions is empty, and we define $\mathfrak{L}$ to be empty, of type 0. On the arithmetic side use the usual absorbing-zero convention: any product with a zero factor is zero. This agrees with successor multiplication and with the eventual limit value after the zero has occurred.

One must not replace this convention by the supremum of *all* earlier partial products after a drop to zero. For instance, a sequence beginning with a positive factor and then a zero has eventual value zero, although the supremum of its entire earlier history is positive. All supremum arguments in this manuscript were explicitly restricted to nonzero factors, where the partial products are nondecreasing.

7.4 What has, and has not, been made nonrecursive

Definitions 3.4 and 7.2 are nonrecursive with respect to Jacobsthal arithmetic. Their proofs of correctness use finite and transfinite induction, just as the correctness proof for the ordinary finite-support construction does. That is not a limit *definition* of the ordered set.

The finite regrouping theorem supplies a direct common-object proof of ordinary binary associativity for $\times_J$. The transfinite model supplies a single ordered set for every arity. It also inherits the known numerical generalized associativity law once its type has been identified. We do not assert that flattening an arbitrarily nested family is automatically an order-preserving map, or give a separate pointwise formula for all such transfinite regrouping isomorphisms.

8 An exact three-state equality test

8.1 States and transitions

At a nonzero partial stage compare the ordinary and Jacobsthal values. Synchronization leaves only three relevant possibilities:

M : equal and monomial, E : equal and nonmonomial, D : different.

In state D neither value can be a monomial, because their supports and leading coefficients agree. Let $q(b)$ be the number of *distinct positive exponents* in the CNF of a nonzero factor b , and let $t = t(b)$.

Theorem 8.1 (Transition table). *Starting from state M at the empty product, a finite list of nonzero factors is classified exactly by Table 1. For an arbitrary ordinal-indexed nonzero list, start from M and read only the finite active suffix. The full ordinary and Jacobsthal products agree exactly when the final state is not D.*

Proof. Let x, y be the synchronized ordinary and Jacobsthal values before the step. If $t = 0$, Lemma 5.1 shows that both new values are $\omega^{d(x)}b$. Strict increase of $\sigma \mapsto d(x) + \sigma$ means their support has exactly $q(b)$ elements. Thus they reset to M when $q(b) = 1$ and to E otherwise.

Suppose $t > 0$. The old exponent support survives in the terminal contribution, and the $q(b)$ new positive-block exponents are distinct and all lie strictly above it. Therefore the new support size is

$$s_{\text{new}} = s_{\text{old}} + q(b). \quad (29)$$

Table 1: Exact transitions after multiplication by a nonzero factor b .

Condition on b	From M	From E	From D
$t = 0, q(b) = 1$	M	M	M
$t = 0, q(b) \geq 2$	E	E	E
$t = 1, q(b) = 0$ (the unit 1)	M	E	D
$t = 1, q(b) \geq 1$	E	E	D
$t \geq 2, q(b) = 0$	M	D	D
$t \geq 2, q(b) \geq 1$	E	D	D

If $x \neq y$, their leading terms agree, so some lower coefficient of y is strictly greater than that of x . That inequality survives and can only grow when all the coefficients of y are multiplied by t . Hence state D persists for $t > 0$.

If $x = y$, the outputs are equal exactly when $t = 1$ or x is a monomial, by the calculation behind Corollary 2.3. Combining this condition with (29) gives all four remaining rows of the table. The initial value 1 is monomial, so the initial state is M.

Finally, Corollary 6.4 reduces equality of arbitrary products to equality on the finite active suffix. The same finite table therefore decides it. Equivalently, the preceding active limit produces an equal pure-power prefix and thus resets the state to M. $\square$

8.2 Interpretation and examples

The automaton has two sources of reset. A nonzero limit factor removes all dependence on previous lower coefficients. A cofinally active limit of stages does the same, and additionally makes the result a pure power. Between resets, discrepancy is created precisely when a finite tail of size at least two multiplies an already nonmonomial equal value. Once discrepancy exists, a factor with positive finite tail cannot remove it.

For example, the active word $(\omega + 1, 2)$ gives $M \rightarrow E \rightarrow D$, so its two products differ. Appending ω gives a reset to M:

$$((\omega + 1) \cdot 2)\omega = \omega^2 = ((\omega + 1) \times_J 2) \times_J \omega.$$

Appending $\omega^2 + \omega$ instead resets to E: the new values agree but have two positive-exponent terms. A later factor 2 creates discrepancy again.

Only the categories of $t(b)$ and $q(b)$ enter the equality decision. The actual ordinal sizes of the exponents and the positive-exponent coefficients do not. This does not mean that the product values are independent of those data; it means precisely that the yes/no comparison with the ordinary product is independent of them once the state is known. The theorem is a finite decision procedure only when the finite active suffix itself is available. It is not an algorithm for discovering that suffix from an arbitrary oracle presentation of an infinite sequence.

9 Worked families and boundary cases

9.1 Three factors with visible cell contributions

Let $a = \omega + 2$. Its square illustrates the distinction between all three multiplications:

$$\begin{aligned} aa &= \omega^2 + \omega \cdot 2 + 2, \\ a \times_J a &= \omega^2 + \omega \cdot 2 + 4, \\ a \otimes a &= \omega^2 + \omega \cdot 4 + 4. \end{aligned}$$

For the direct triple product, a pivot at coordinate 3 contributes one cell of type ω^3 . A pivot at coordinate 2 contributes two cells of type ω^2 , one for each terminal point at coordinate 3. A pivot at coordinate 1 contributes four ω -cells and eight singletons. Thus

$$F(a, a, a) = \omega^3 + \omega^2 \cdot 2 + \omega \cdot 4 + 8. \quad (30)$$

The ordinary cube is $\omega^3 + \omega^2 \cdot 2 + \omega \cdot 2 + 2$. The support and leading monomial agree, while the lower coefficients differ exactly as the synchronization theorem permits.

9.2 A closed form for $(\omega + c)^{\times_J n}$

For $c \geq 1$ finite and $n \geq 1$, write $J_n(c)$ for the product of n copies of $\omega + c$. The cell formula immediately gives

$$J_n(c) = \sum_{j=n}^0 \omega^j c^{n-j}. \quad (31)$$

In contrast,

$$\begin{aligned} (\omega + c)^n &= \omega^n + \sum_{j=n-1}^0 \omega^j c, \\ \underbrace{(\omega + c) \otimes \cdots \otimes (\omega + c)}_{n \text{ factors}} &= \sum_{j=n}^0 \omega^j \binom{n}{j} c^{n-j}. \end{aligned} \quad (32)$$

Every descending summation here means ordinary CNF concatenation in the displayed decreasing order of exponents. Thus the formulas also apply without ambiguous ellipses when $n = 1$.

To verify (31) directly, the final pivot contributes ω^n , the preceding pivot contributes c copies at exponent $n - 1$, and so on. The finite-only tuples contribute c^n singletons. The ordinary formula follows by induction using (7). The natural formula is the finite binomial theorem for the natural semiring; its exponents here are finite, so natural and ordinary addition of exponents agree.

These are applications of the construction, not claims that the numerical finite-power identities were previously unknown. They also show that the coefficient differences can be very large while $O_n \leq J_n < O_n \cdot 2$ still holds. An ordinal-scale bound is not a coefficientwise ratio bound.

9.3 Passing through a genuine limit and then taking successors

For the constant sequence $a = \omega + 2$ indexed by ω , all indices are active. The common prefix product is ω^ω . At index $\omega + 2$, the last two factors constitute the finite active suffix. Hence

$$\begin{aligned} J_{\omega+2} &= \omega^{\omega+2} + \omega^{\omega+1} \cdot 2 + \omega^\omega \cdot 4, \\ O_{\omega+2} &= \omega^{\omega+2} + \omega^{\omega+1} \cdot 2 + \omega^\omega \cdot 2. \end{aligned}$$

Discrepancy has disappeared at ω and reappeared after two successor factors. The automaton reads only $(\omega + 2, \omega + 2)$ and ends in D.

More generally, for any $a > 1$ and $\beta = \lambda + n$ with $\lambda = 0$ or a nonzero limit,

$$a^{\times_J \beta} = a^\lambda F(\underbrace{a, \dots, a}_{n \text{ copies}}). \quad (33)$$

This is Jacobsthal exponentiation, the iteration of $\times_J$. It is not super-Jacobsthal exponentiation, which iterates $\otimes$. The direct finite-support construction therefore also supplies a model for the operation in (33).

9.4 Exponent absorption and the separation of pivot layers

Suppose the degrees of successive factors include ω and ω . Their ordinary sum is $\omega \cdot 2$, not a polynomial operation on integer degrees. Suppose instead a prefix degree is finite and a later pivot exponent is ω ; that finite prefix is absorbed: $k + \omega = \omega$. Such absorption affects the calculated type and must be handled correctly in any implementation of the cell formula.

It does *not*, however, cause a collision between different pivot layers. Lemma 3.3 shows that a pivot- j exponent lies above all preceding prefix degrees and at most D_j ; all later pivot layers lie strictly higher. Within one pivot layer, distinct input block exponents remain distinct after addition of the common prefix degree. Equal cell exponents therefore arise only from repeated blocks of the same exponent and their different finite tail choices.

This separation explains why the primitive comparison rule needs no prefix arithmetic at all. It also prevents a tempting but incorrect concern that ordinary exponent absorption might reverse the order of pivots. Regrouping still requires matching ordered blocks rather than raw coordinate tags: the inner colex orders change under grouping, as Remark 4.3 demonstrates.

9.5 Why an unrestricted infinite maximum is the wrong target

The componentwise order on finite-support functions from ω to 2 has an infinite antichain: the functions supported at a single coordinate. Thus the finite Cartesian-product maximal-order theorem cannot simply be applied to the full finite-support product. Lipparini discusses this obstruction and the need for additional restrictions in [5, Section 3.4].

Our transfinite maximality statement deliberately fixes the ordinary order within its finitely many large prefix-suffix cells. It does not claim that Jacobsthal multiplication, or infinitary natural multiplication, is the maximum over all linear extensions of the unrestricted componentwise finite-support order.

10 Reproducible checks and their limitations

10.1 The exact ordinal kernel

The archive contains a standard-library-only Python implementation of hereditary Cantor normal forms below ε_0 . An ordinal is an immutable finite tuple of strictly decreasing exponent–coefficient pairs; each exponent is itself an ordinal in the same representation. Zero is the empty tuple. The implementation provides ordinary addition and multiplication, natural addition and multiplication, the classical binary Jacobsthal formula, and a separate implementation of the all-arity pivot formula.

The pivot routine never invokes the binary Jacobsthal routine. It computes all ordinary prefix degrees, all finite suffix-tail products, and the resulting cell monomials, then combines equal exponents. This provides a useful comparison between independently organized formulas, although they share the same low-level ordinal kernel.

The actual tuple comparison rule is also implemented for sampled points in canonical blocks. A point records its block index, block exponent, offset within the block, and position in the factor. This allows checks of the componentwise-extension property independently of the type formula. A second comparator uses the calculated cell degrees; it is checked against the primitive, prefix-arithmetic-free rule on every tested pair.

10.2 Test families and recorded results

The test run in the archive used random seed 20260920. All assertions passed. Table 2 records the executed counts, rather than planned counts.

Table 2: Exact checks recorded in `results/verification_results.json`.

Test family	Count
Binary pairs, including zero, over the small CNF domain	729
Exhaustive triples over 26 positive polynomials below ω^3	17,576
Additional finite lists with nested transfinite exponents	3,500
Finite split/regrouping comparisons	86,216
Explicit lists containing a zero factor	15
Explicit empty-list case	1
Sampled or exhaustively enumerated tuple-pair comparisons	56,769
Agreements of primitive and cell-degree comparison rules	56,769
Componentwise-comparable pairs within that set	12,107
Additional tuple-triple transitivity checks	4,000

The small domain contains all nonzero expressions $\omega^2 c_2 + \omega c_1 + c_0$ with $c_i \in \{0, 1, 2\}$, giving $3^3 - 1 = 26$ values. The additional lists have lengths from zero to seven and use exponents selected from a pool containing $0, 1, 2, 3, \omega, \omega + 1, \omega \cdot 2, \omega^2, \omega^\omega$, and $\omega^\omega + \omega$. Positive coefficients range from one to four.

For every nonzero list, the test checks the direct formula against binary iteration; all finite split points; equality of supports and leading monomials; coefficient domination; the strict factor-two bound; the degree and finite-tail formulas; the natural-product upper bound; and the three-state classification. Binary tests also check the defining successor equation and the exact binary equality criterion.

The tuple tests check comparison antisymmetry, separation of distinct points, componentwise extension, and sampled transitivity. Some small families are exhausted, but only among the selected points; an infinite block is never replaced by a claim that its finite sample is exhaustive.

10.3 What the tests cannot establish

The computational domain is below ε_0 , whereas the theorems are stated for all ordinals. No arbitrary transfinite limit is evaluated by the program. The displayed $\omega + 2$ examples are computed *after* substituting the prefix value justified by the proof. In particular, random tests do not certify the limit argument, the pure-power theorem, or the correctness of the reduction for arbitrary cofinalities.

The implementation is not a formalization in Lean, Isabelle, or another proof assistant. Sharing the low-level CNF routines creates a possibility of correlated implementation mistakes. The checks should be viewed as structured attempts to falsify finite consequences and catch transcription errors, not as substitutes for the English proofs.

11 Assessment, scope, and remaining questions

11.1 The precise positive result

The manuscript gives an affirmative construction for the following precise version of the selected question: there is an isomorphism-invariant, CNF-based well-order on finite tuples, and on finite-support tuples at arbitrary ordinal arities, defined without Jacobsthal iteration or its limit value, whose type is the Jacobsthal product. The finite objects prove associativity through an explicit comparison of their blocks. The restricted maximality theorems describe the same values without choosing a tie-breaking order.

The binary comparison with natural multiplication also has the requested order-theoretic explanation: the constructed order is one componentwise linear extension, while the natural product is the maximum among all such extensions. This addresses the binary inequality part of Question 5.2, not every question listed there.

These statements are stronger than exhibiting a CNF formula for a parenthesized product: the underlying sets, cells, comparison rules, componentwise compatibility, and regrouping argument have all been specified. On the other hand, the adjective “natural” in the original question has no unique formal definition. The construction uses canonical Cantor blocks and should be judged against that explicit design choice.

11.2 Claims not made

We do not claim the classical associativity law, the classical binary CNF formula, the ordinary finite-support representation, or the numerical inequalities as new. We do not claim that our comparison rule is preserved by arbitrary embeddings, that it is independent of canonical CNF structure, or that the tuple-flattening map is the associator. We do not settle Altman’s separate questions about super-Jacobsthal exponentiation or all infinitary natural-product inequalities.

The literature search did not establish a priority theorem. In particular, the synchronization and finite-tail observations may have antecedents not found by the targeted search. The contribution presented for evaluation is the combination of the concrete pivot model, its finite-cell extremal

description, its blockwise associativity proof, and the arbitrary-index finite-tail representation, with all dependencies made explicit.

11.3 Further research directions suggested by the construction

A useful next mathematical question is whether the pivot-cell constraints can be replaced by an intrinsic condition on order embeddings, convex suborders, or initial segments, without first displaying the CNF blocks. Such a characterization would clarify how canonical this model is among other possible answers to the same informal question.

A second question is how to describe the unique regrouping isomorphisms by explicit coordinate transformations. The example in Remark 4.3 shows that coordinate flattening is insufficient; some absorption isomorphisms inside monomial cells are unavoidable. An effective description for hereditary CNFs could be suitable for formalization.

A third direction is to identify the exact relationship between the finite-suffix maximality condition here and the finite-prefix conditions used for countable natural products. Their orientations differ: the Jacobsthal discrepancy is confined to the *end* of the active index, whereas the countable natural-product construction in [5] uses a finite initial part followed by ordinary infinite behavior. This comparison is a research direction, not a claimed new equivalence.

12 Conclusion

The main obstruction to a direct infinitary Jacobsthal model turns out, in this approach, not to be an infinite combinatorial object of a new kind. The relevant infinite object is the ordinary finite-support product. What is different is a finite terminal rearrangement, organized by the last coordinate occupying a positive-exponent Cantor block.

The finite partition makes the all-arity object explicit. The common leading monomial bounds every discrepancy by the next ordinary doubling. Cofinal activity then forces the two operations to agree at a limit, leaving only finitely many active successor factors to be handled by the pivot construction. Together these observations yield a concrete proposed answer to the selected open-ended order-theoretic question, with a clear account of both its mathematical strength and its limitations.

A Proof dependency map and audit checklist

The mathematical dependency chain is as follows. Ordinary CNF absorption proves the classical binary formula directly from the defining recursion. The pivot partition and ordinary cell orders give their monomial types. The explicit finite cell formula then proves the binary identification, the metadata lemma, and the finite regrouping theorem. None of these steps invokes Jacobsthal associativity as a premise.

Separately, the binary formula gives the one-step synchronization lemma. That lemma and the factor-two estimate are proved simultaneously through all ordinal stages. At a cofinally active limit the estimate yields mutual cofinal domination, hence equality; additive indecomposability then yields a pure power. Unit erasure and finite associativity produce the finite-tail reduction. The direct ordinary finite-support model, together with the finite pivot model, gives the final all-at-once

order.

The three most delicate checks are worth isolating. First, the global pivot can only move left under a coordinatewise increase, and such a move strictly lowers its cell exponent. Second, at a cofinally active limit, for each earlier stage ξ there really is a later successor stage $i + 1 < \lambda$ that overtakes J_ξ ; the limit hypothesis ensures this successor is still below the endpoint. Third, the active index, not the ambient ordinal index, controls the finite suffix. Example 5.5 shows exactly what would go wrong if units were not removed.

The mathematical argument is not restricted to countable ordinals. Every invocation of cofinality chooses a later active index individually; no countable cofinal sequence is assumed. The CNF decomposition has finitely many terms for each input ordinal, regardless of its cardinality. The number of factors may be any set-sized ordinal, but proper-class products are outside the scope of the construction.

B Using and rebuilding the archive

The principal files are `article.tex`, `article.pdf`, `code/ordinals.py`, and `code/verify.py`. The `results` directory contains machine-readable and plain-text records of the executed checks. The `notes` directory contains a short proof audit and a source-search log. No downloaded third-party article PDFs or font files are redistributed.

From the archive's top-level directory, run

```
python3 code/verify.py
latexmk -pdf -interaction=nonstopmode -halt-on-error article.tex
```

The supplied `Makefile` provides `make check`, `make pdf`, and `make clean`. The source has an internal bibliography, so no separate bibliography processor is required. The Python checks require only the standard library. The LaTeX document uses common packages included in standard TeX Live installations.

References

- [1] H. Altman, *Intermediate arithmetic operations on ordinal numbers*, Mathematical Logic Quarterly **63** (2017), 228–242. DOI: [10.1002/malq.201600006](https://doi.org/10.1002/malq.201600006). Author version: [arXiv:1501.05747](https://arxiv.org/abs/1501.05747), version 9, 18 September 2017. Questions 2.16 and 5.2 are the selected problem references.
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- [3] P. W. Carruth, *Arithmetic of ordinals with applications to the theory of ordered Abelian groups*, Bulletin of the American Mathematical Society **48** (1942), 262–271. [Project Euclid record](#). The finite-product maximum theorem is also explicitly recalled in [5, Section 3.3].
- [4] P. Lipparini, *Some transfinite natural sums*, Mathematical Logic Quarterly **64** (2018), 514–528. [arXiv:1509.04078](https://arxiv.org/abs/1509.04078). The earlier title cited by Altman was *Some iterated natural sums*.
- [5] P. Lipparini, *An infinite natural product*, Annales Mathematicae Silesianae **32**(1) (2018), 247–262. [arXiv:1703.06908](https://arxiv.org/abs/1703.06908), version 2. Sections 3.2–3.4 describe the ordinary finite-support product, the finite natural-product maximum theorem, and the obstruction to an unrestricted infinitary extension.
- [6] P. Lipparini, *Noncommutative infinitary semigroups*, [arXiv:2605.28413](https://arxiv.org/abs/2605.28413), version 1, 27 May 2026. Cited as related general context, not as a source of the Jacobsthal construction proposed here.